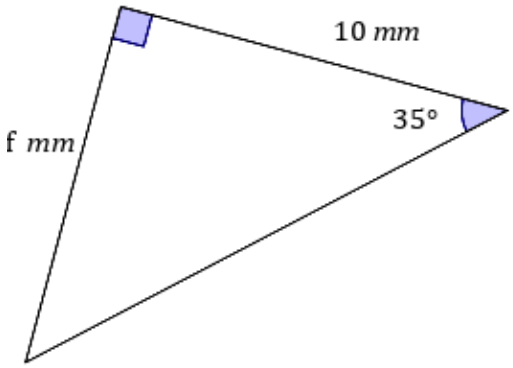
Solving for sides in right triangles

1 Consider the diagram shown.



a Is side with length f mm adjacent or opposite the 35° angle?

b Find the length of the side adjacent to the 35° angle.

c Should the length of the side opposite to the 35° angle be greater than or less than 10 mm?

d Determine whether the following equations involving f are true.

i $f = \dfrac{\tan 35^\circ}{10}$

ii $f = \dfrac{10}{\tan 35^\circ}$

iii $\tan 35^\circ = \dfrac{10}{f}$

iv $\tan 35^\circ = \dfrac{f}{10}$

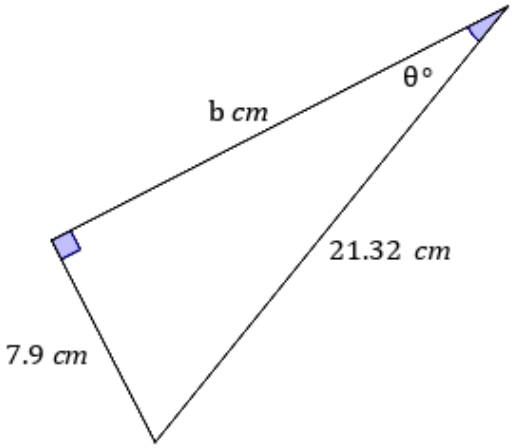
v $f = 10 \tan 35^\circ$

e Hence, solve for f, correct to one decimal place.

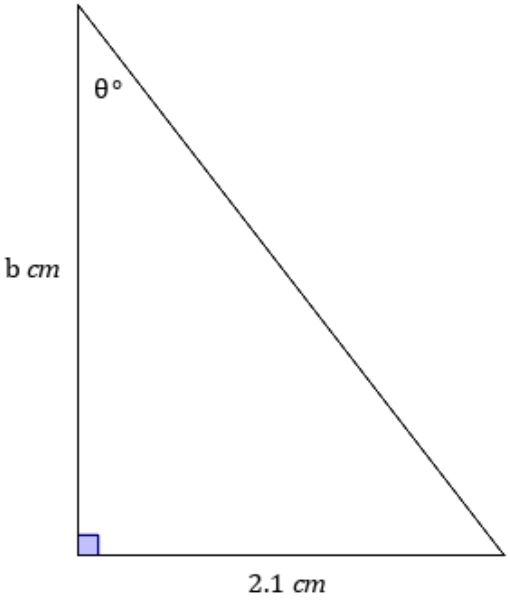
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2 Given the value of $\tan\theta$, find the value of b on the following diagrams. Round your answers to one decimal place when necessary.

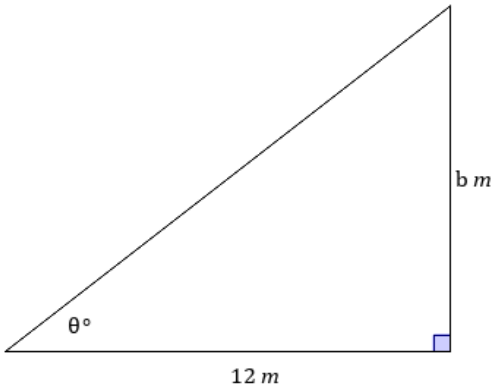
a $\tan\theta = 0.4$



b $\tan\theta = 0.3$



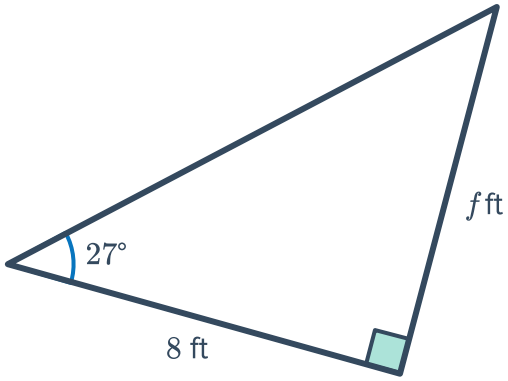
c $\tan\theta = \frac{2}{3}$



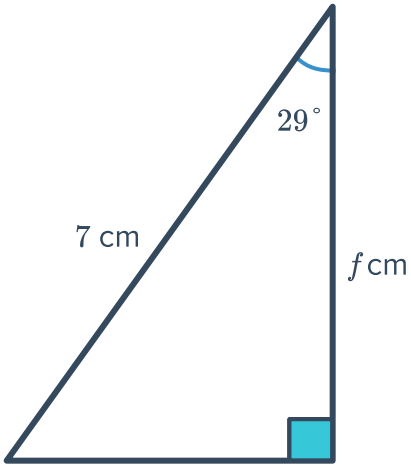
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3 Find the value of the variables in the following triangles, rounding your answer to two decimal places when necessary.

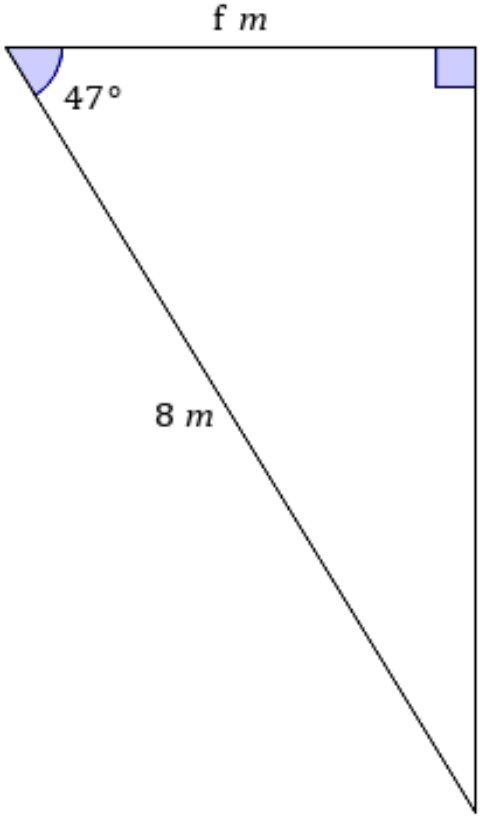
a



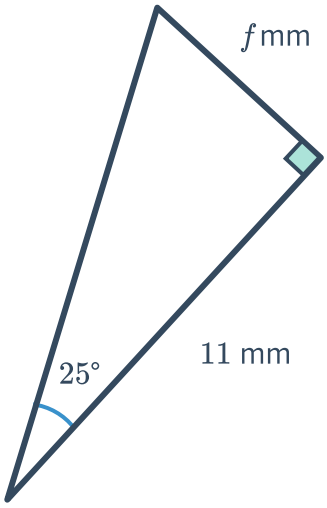
b



c

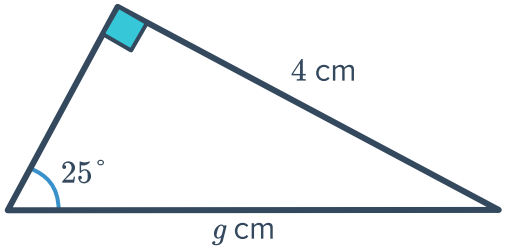


d

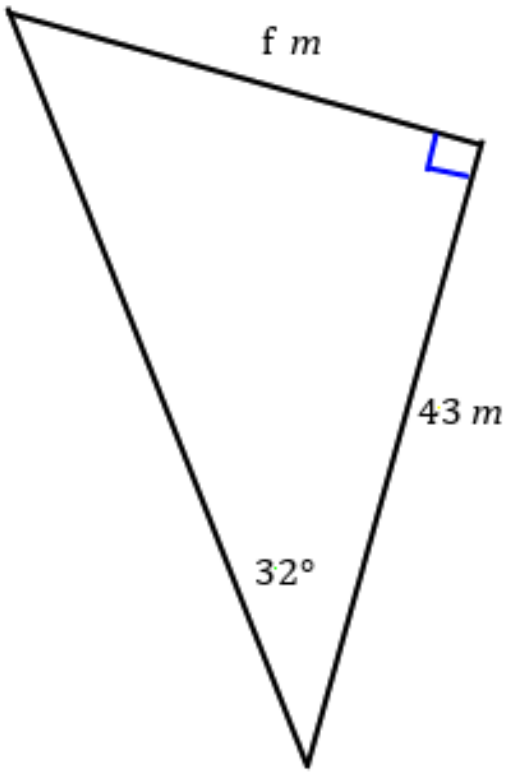


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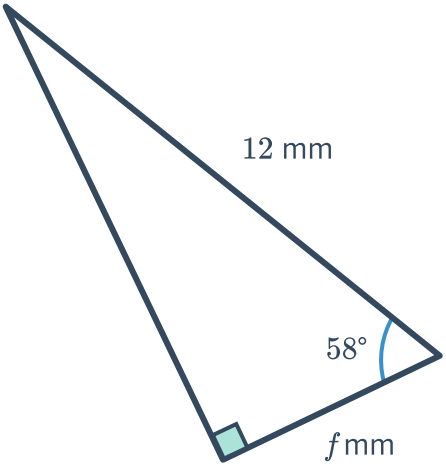
e



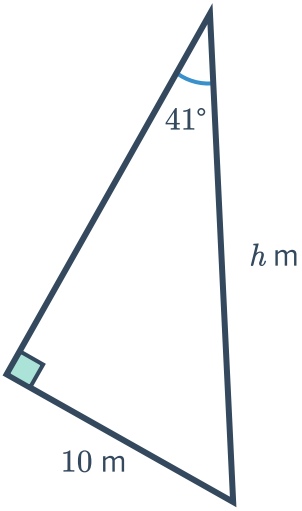
f



g

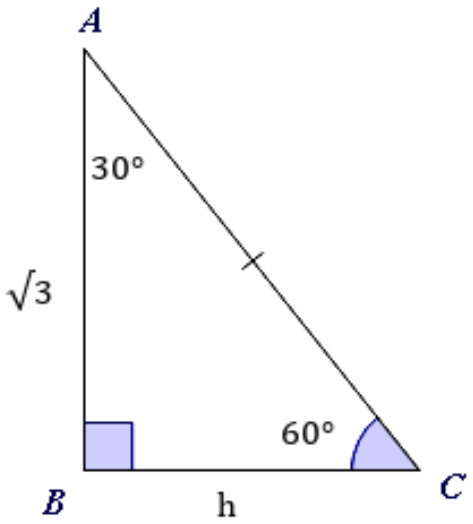


h

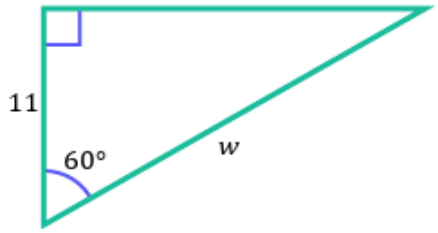


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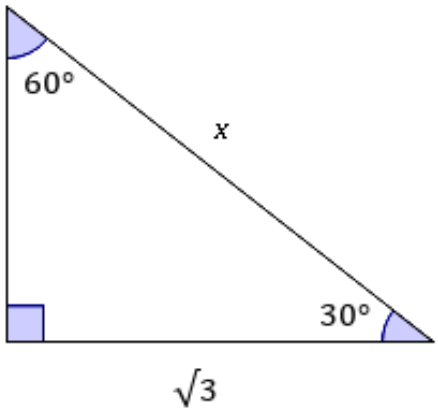
i



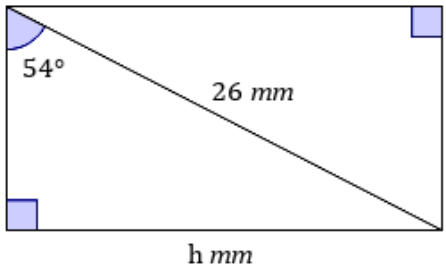
j



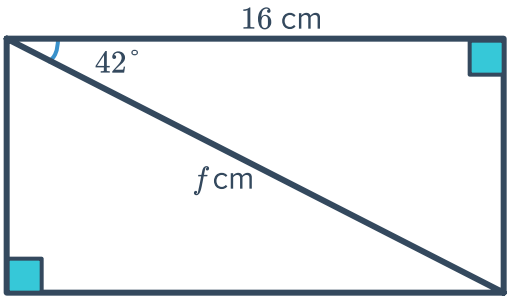
k



l

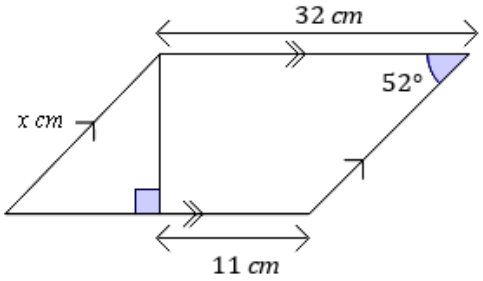


m

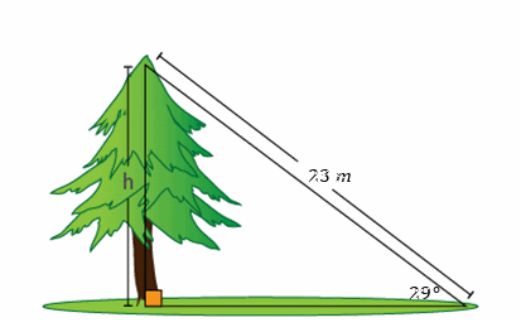


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4 Consider the given diagram.
Find the value of x, the side length of the parallelogram. Round your answer to the nearest centimeter.



5 Consider the following figure.
If the height of the tree is h meter, find h to two decimal places.

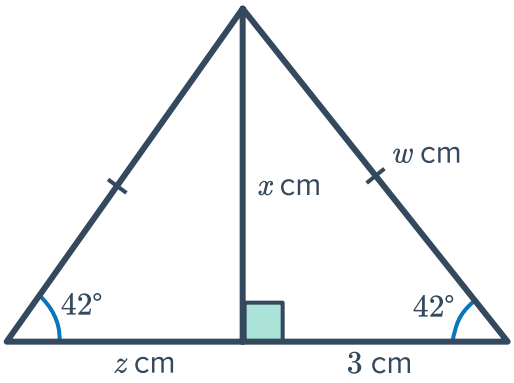


6 Consider the following diagram.

a Find the value of x, rounding your answer to two decimal places.

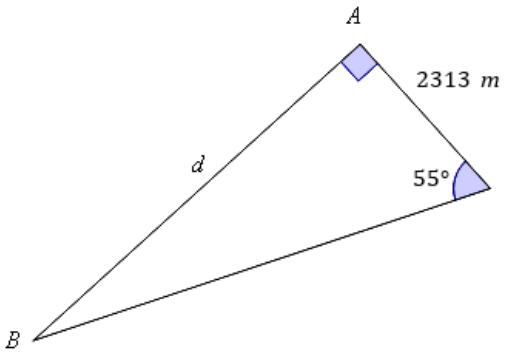
b Find the value of w, rounding your answer to one decimal place.

c Find the value of z, rounding your answer to the nearest whole number.

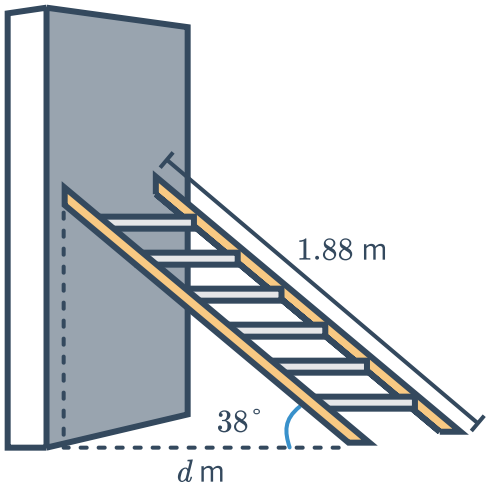


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7 Neil's home is located at point A, and Neil is on his skateboard at point B.
If d is the distance between Neil and his home, calculate d, correct to two decimal places.

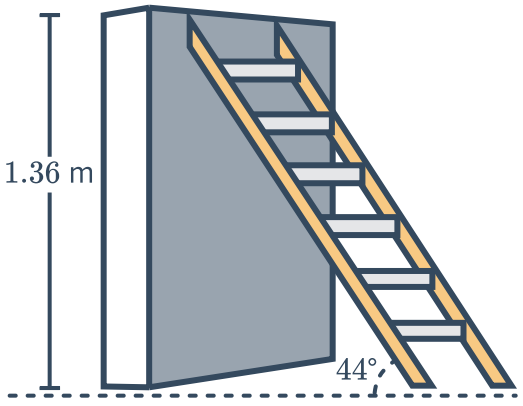


8 Consider the given figure.
If d is the distance between the base of the wall and the base of the ladder, find d to two decimal places.



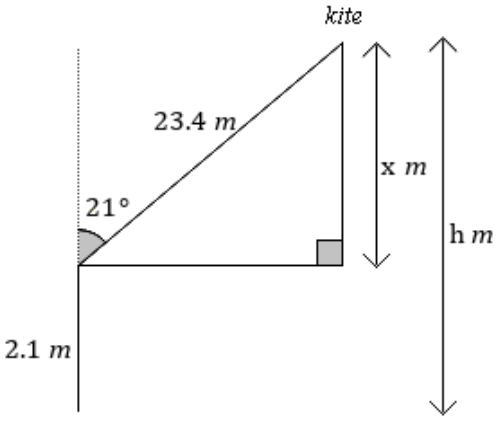
9 The longer side of a rectangular garden measures 14 m. A diagonal path makes an angle of 26° with the longer side of the garden.
If the length of the shorter side of the garden is y m, calculate y to two decimal places.

10 Consider the following figure.
If l is the length of the ladder in meters, find l, to two decimal places.



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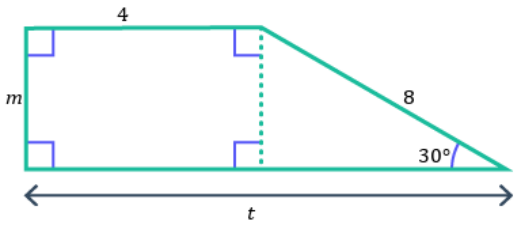
11 A girl is flying a kite that is attached to the end of a $23.4\ \text{m}$ length of string. The angle between the string and the vertical is 21°. If the girl is holding the string $2.1\ \text{m}$ above the ground:



a Find x, correct to two decimal places.

b Hence find the height h of the kite above the ground, correct to two decimal places.

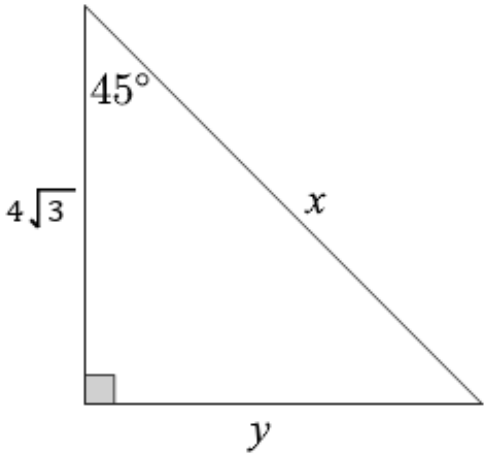
12 Consider the unknown lengths in the given diagram.



a Find exact value of m.

b Find exact value of t.

13 Consider the adjacent diagram.

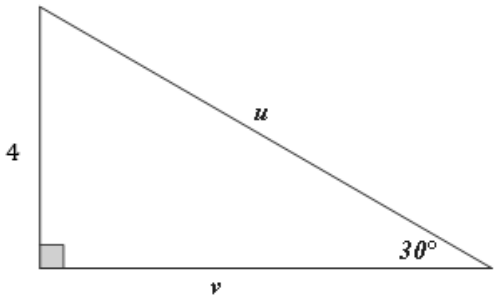


a Solve for the exact value of x.

b Solve for the exact value of y.

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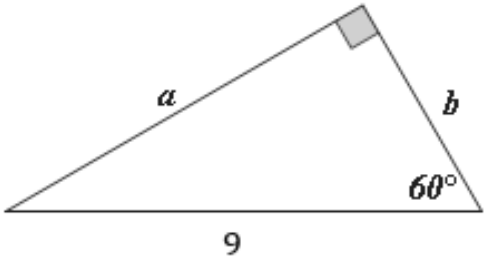
14 Consider the given diagram.



a Solve for the exact value of v.

b Solve for the exact value of u.

15 Consider the adjacent diagram.



a Solve the exact value of a.

b Solve the exact value of b.

Additional questions

16 Evaluate the following, rounding your answer to two decimal places:

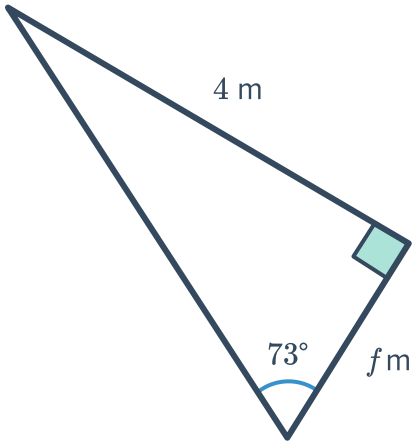
a $\sin 82^\circ$ **b** $\cos 28^\circ$ **c** $\tan 84^\circ$ **d** $9 \sin 63^\circ$

e $7 \cos 77^\circ$ **f** $8 \tan 28^\circ$ **g** $\dfrac{\sin 71^\circ}{7}$ **h** $\dfrac{\cos 7^\circ}{9}$

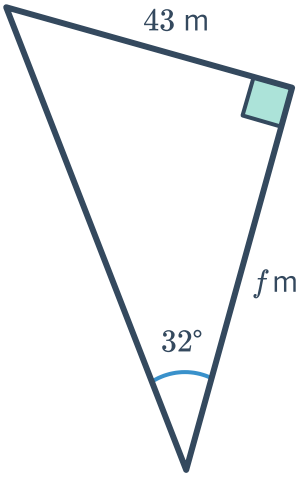
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17 Find the value of the variable in the following triangles, rounding your answer to two decimal places:

a

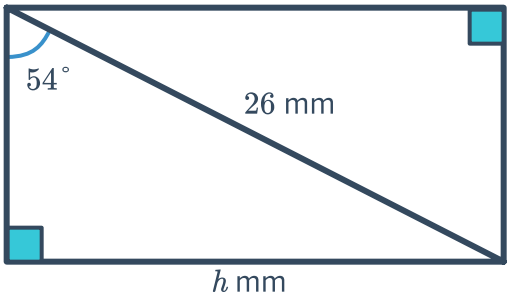


b

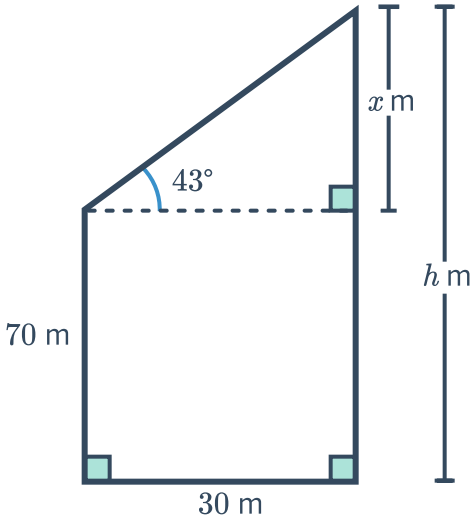


18 Find the value of the variable(s) in the following diagrams, rounding your answer to two decimal places:

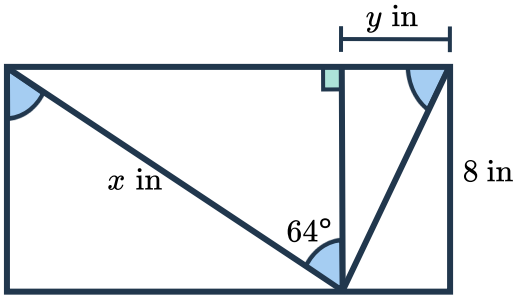
a



b



c

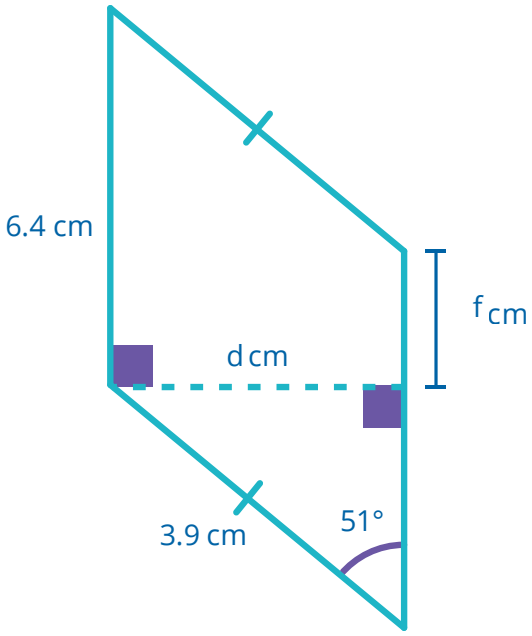


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19 Consider the shape given.

a Find the value of d, rounding your answer to one decimal place.

b Find the value of f, rounding your answer to one decimal place.



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